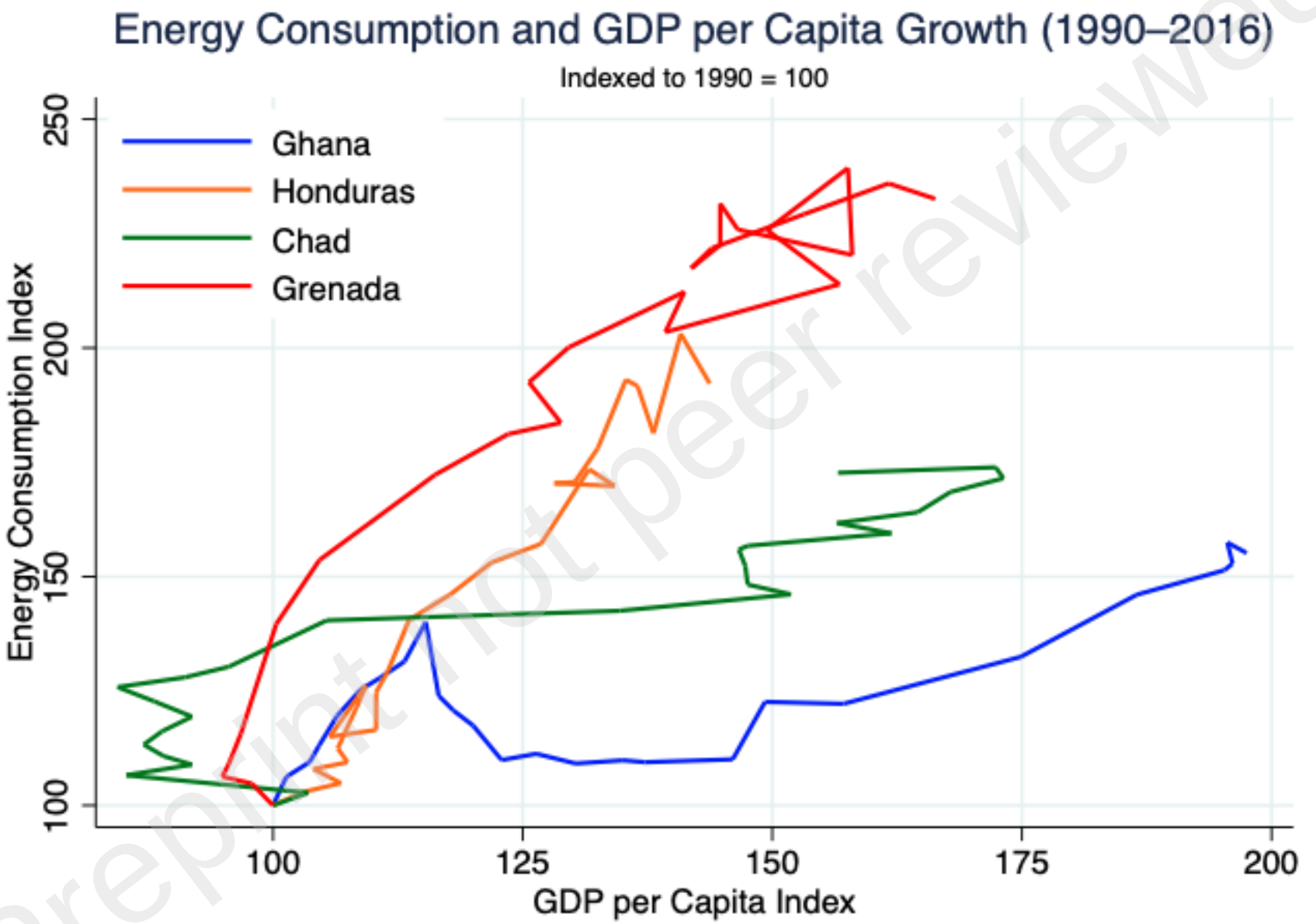


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Energy efficiency and economic growth: Evidence from low-income countries

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Energy efficiency and economic growth: Evidence from low-income countries

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Abstract

This paper contributes to the understanding of the relationship between energy efficiency (EE) and economic growth in 28 low-income countries (LICs) from 1991 to 2016. I provide empirical evidence that highlight heterogeneities often overlooked in prior literature. Using second-generation panel techniques, I estimate both short- and long-run effects while controlling for energy prices. My results show that EE significantly contributes to long-run growth but exerts negative short-run impacts due to transitional frictions such as infrastructure gaps, capital constraints, and institutional rigidity. In addition, Cluster-specific analyses show differing results: wealthy, high-efficiency LICs experience decreasing returns; poorer, low-efficiency LICs face growth-reducing effects; and structurally unique economies show no significant long-term gains. Thus, this implies the need for country-specific, sector-targeted policies for unlocking the growth potentials of EE.

Keywords: Energy efficiency, Economic growth, Low-income countries, Cross-Sectionally Augmented ARDL, Error-correction, Cointegration.

JEL classification: O1, O4, O5, Q4.

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1 Introduction

Many countries embrace energy efficiency (EE) to reduce emissions and foster sustained growth. In high and middle-income countries, empirical evidence shows that efficiency improvements often enhance productivity, stimulate innovation, and support long-term economic growth (see e.g., Chen et al., 2024; DellAnna, 2021; Santos et al., 2024; Ahmed and Elfaki, 2024; Deka et al., 2023; Xu et al., 2024; Brlei Vali and Markuz, 2024; Ayres & Warr, 2009; Howarth, 2007). However, in low-income countries (LICs), the relationships are less clear because these countries face several constraints that complicate the EE-growth nexus in addition to concerns about the trade-off between energy efficiency and economic growth.This is particularly the case due to low-income levels and limited fiscal capacity, yet a large increase in energy production and consumption is needed to keep up with the expansion of economic activities. These differences make it difficult to generalize from existing studies and underscores the need for evidence on whether, and through which channels, efficiency improvements influence growth in LICs.

Low-income countries differ widely in terms of institutional strength, economic power and structure, energy systems and efficiency, resource endowment, and, electricity generation and distribution capacity. For instance, LICs such as the Maldives, Ghana, Cabo Verde, or Senegal., possess the infrastructure and governance frameworks to support energy-efficient investments, whereas Burkina Faso, Mali, Chad, Congo, Honduras, or Togo., face deep structural constraints such as limited capital, weak regulatory institutions, or a heavy reliance on traditional energy sources.[1] Similarly, their electricity generation and access range from reliance on hydropower, fossil fuels, coal, renewable, or biomass to contexts marked by limited grid infrastructure and pervasive energy poverty- factors that can fundamentally alter the transmission mechanisms and impact of EE investments.[2]Despite these challanges, international financial support assist these countries to finance efficiency investments.[3]Given these heterogeneities, I study the effects of EE on LICs, allowing for a more precise evaluation of when, where, and how energy efficiency can serve as an effective factor shaping economic growth.

My main goal is to analyze the overall effect of EE on growth and empirically identify heterogeneous

[1]Relatively wealthier LICs such as Maldives, Ghana, Cabo Verde, and Senegal have stronger infrastructure, diversified economies, electrification rates near 90%, GDP per capita around $7000, and more capable institutions, allowing them to translate EE gains into long-term productivity. In contrast, Burkina Faso, Mali, Chad, and Togo remain heavily reliant on biomass or traditional fuels, have electrification rates near 25%, GDP per capita around $2500, and face capital and institutional constraints that limit the growth impact of EE investments.

[2]Electricity generation capacity varies widely across LICs, shaping the impact of EE initiatives. Cabo Verde produces about 506 GWh, with 72% from fossil fuels and 28% renewables, reflecting small scale and import dependence. Senegal generates about 6,654 GWh from a diversified mix of thermal (73%), solar (13%), wind (8%), and hydro (6%). Burkina Faso produces only 1,697 GWh, mostly from fossil fuels and with high import reliance, indicating weak infrastructure. Ghana stands out with more than 23,000 GWh from a balanced mix of hydro (3439%), thermal (6065%), and emerging renewables, allowing for stronger efficiency gains

[3]These international financial flows come from official development assistance (ODA) and other official flows (OOF) from donor governments, development financial institutions, export credits, and multilateral agencies. They cover loans, grants, and equity investments directed to clean energy research, development, and production, as well as renewable or hybrid energy systems, technical assistance, and renewable electricity distribution infrastructure

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features. Specifically, I ask three research questions. First, do improvements in EE contribute to GDP growth in LICs? Second, are there distinguishable short-run and long-run effects of EE on economic growth? Third, what unique country-specific conditions shape the EE-growth nexus? To answer these questions, I collect data on energy efficiency, energy prices, and gross domestic product (GDP) for 28 LICs countries from 1991 to 2016.[4]These data are drawn from the World Banks World Development Indicators and energy price data from the World Banks inflation database developed by Ha, Kose, and Ohnsorge (2023).To account for heterogeneities, I adapt the International Monetary Funds LIC classification by grouping these countries into four distinct clusters based on key economic and energy-related metrics: (i) wealth, (ii) energy efficiency, (iii) resource richness, and (iv) economic uniqueness. This classification allows me to capture the different effects of EE under diverse conditions.

My empirical strategy follows four stages, in line with Deka et al. (2023) and Xu et al. (2024). First, I test for cross-sectional dependence across countries using Pesaran (2015) to account for global linkages and shared vulnerabilities. Second, panel unit root tests (CIPS) assess the integration properties of the data, followed by Pedroni and Westerlund cointegration tests to confirm the existence of long-run relationships among GDP, EE, and energy prices. Finally, the main estimation technique is a Cross-Sectionally Augmented Autoregressive Distributed Lag (CS-ARDL) model using the Pooled Mean Group (PMG) estimator, which captures heterogeneous short-run dynamics and imposes homogeneity in long-run coefficients. This method accommodates different speeds of adjustment, controls for both cross-country spillovers, endogeneity, and serial correlations. In addition, it produces consistent estimates and avoids the potential bias associated with unit root and cointegration tests.

My empirical results show that overall a unit change in EE significantly contributes to positive economic growth in the log-run, but exerts a negative and statistically significant short-run effect. In other words, while efficiency gains eventually support growth through improvements in productivity and innovation, they initially impose adjustment costs that weigh on short-run gains. This finding corroborates prior literature using data from high and midddle-income countries to contribute to the understanding of the relationship between energy efficiency and growth (see e.g., Chen et al., 2024; DellAnna, 2021; Santos et al., 2024; Ahmed and Elfaki, 2024; Deka et al., 2023; Xu et al., 2024; Brlei Vali and Markuz, 2024; Dzwigol et al., 2023)[5] [6]

[4]Cluster 1 (Wealthy and Efficient): Cabo Verde, Ghana, Kenya, Maldives, Senegal, Zambia. Cluster 2 (Resource-rich economies): Cameroon, Congo, Côte d'Ivoire, Guinea, Honduras, Kyrgyzstan. Cluster 3 (Low Efficiency and Limited Wealth): Benin, Burkina Faso, Burundi, Chad, Mali, Mozambique, Togo, Lesotho. Cluster 4 (Unique Economic Structures or Characteristics): Ethiopia, Myanmar, Nepal, Grenada, Saint Lucia, St. Vincent and the Grenadines, Tajikistan, Tonga, Zambia

[5]Akram et al.(2021), Deichmann et al.(2018), Oyewole et al. (2024), Ziolo et al. (2020), Alola et al. (2023), Hori(2012), Sueyoshi and Goto(2023), Liu and Chen (2025), and Lin and Zhou(2020) examine the impact of energy efficiency on economic growth, reporting heterogeneous effects like mine but shaped by factors such as development stage, province or region, income group, continental location, or membership in economic blocs. However, none of these studies investigate the underlying heterogeneitiessuch as economic structure, energy systems and efficiency, and resource capabilitieswithin countries, nor how these factors condition the EEgrowth relationship

[6]In addition, Adom et al. (2021) examined the moderating role of income inequality in shaping the relationship

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Next, the cluster-specific analyses, the results show that in wealthy high-efficiency LICs, a unit change in EE leads to positive but insignificant long-run effects likely due to diminishing returns from existing efficiency gains (a characteristic common to high-income countries). By contrast, lower-efficiency and poorer LICs face negative growth effects from efficiency initiatives in both the short and long run. This could be attributed to structural constraints, weak absorptive capacity, adoption barriers, capital limitations, and institutional weaknesses. For the final group of countries, those with structurally unique economies, such as their dependence on tourism or agriculture(for example, Myanmar, Grenada, Nepal, Tajikistan, or Tonga), do not show a significant long-term relationship between EE and growth, indicating that energy efficiency is not a primary growth driver in these countries. Specifically, a 1 % increase in EE leads to 7 % increase in economic growth. These findings align with previous studies and highlight the conditional nature of the EEgrowth relationship, underscoring the importance of tailoring strategies to country-specific contexts rather than relying on one-size-fits-all policies

Previous studies, such as Chen et al., (2024); Cantore et al. (2016); Akram et al.(2021), and Dzwigol et al. (2023), focus on high- and middle-income countries, and establish generally positive or bidirectional relationships between EE and growth. In contrast, the literature on LICs such as (2021), Rajbhandari and Zhang (2018), (2021) Fowlie and Meeks, and Adom et al. (2021) remains sparse, often treating them as homogeneous groups. Notably, Fowlie and Meeks (2021) provide strong evidence that while energy efficiency has the potential to support economic growth in the middle and low-income countries, its effectiveness is often undermined by barriers such as income inequality, weak institutions, limited access to credit, poor infrastructure, unreliable power supply, and governance failures. In this literature, I contribute to the understanding of the relationship between EE and economic growth in two ways. First, I focus exclusively on LICs rather than aggregating them with wealthier economies. Second, I introduce a novel four-cluster framework based on wealth, energy efficiency, resource endowment, and economic structure leveraging the IMF classifications, which allows for a better understanding of the EE growth relationship. As a result, my study reveals distinct relationships and policy implications of energy efficiency that prior studies overlook. In particular, it highlights that in poorer LICs such as Burundi and Chad, EE investments often impose lasting transitional costs without leading to long-run productivity gains, while in economies dominated by low energy-intensive sectorssuch as tourism or subsistence agriculturethe aggregate growth effect of EE remains negligible, even when sector-specific improvements are observed. Hence, this study offers a more context-sensitive foundation for policy design in low-income countries.

The remainder of this paper is organized as follows. In the next section, I discuss the empirical strategy, while section 3 describes the dataset and clustering method. Sections 4 and 5 analyzes the

between energy efficiency and economic growth. In contrast, and in complete divergence from my findings, Sinha (2011) reports a unidirectional causal link running from GDP to reductions in energy wasteboth in the short and long runsuggesting that, in the Indian context, energy efficiency is primarily an outcome of economic growth rather than a driver of it

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results, and evaluates the studys contribution respectively. Finally, Section 6 concludes the paper.

2 Methodology

I employ second-generation panel techniques to examine energy efficiency's dynamic impact on economic growth in 28 low-income countries, addressing cross-sectional dependence and heterogeneous dynamics. This is important because ignoring these cross-country dependencies and linkages would lead to biased or inefficient estimates. Similar methodology has been applied by Xu et al. (2023), Deka et al. (2023), Ahmed and Elfaki (2024), and Yosthongngam and Tansuchat (2023) in their research on OECD countries, emerging economies, Asian middle-income countries, and G7 countries. My analysis is similar to these studies in applying second-generation panel methods to capture both long-run equilibrium relationships and heterogeneous short-run dynamics. However, it differs in its exclusive focus on low-income countries, the introduction of a four-cluster classification to account for the prevailing heterogeneities, and the explicit account of energy prices to capture market influences.

Departing from prior research, which focused predominantly on advanced or middle-income economies, I implement a comprehensive four-step second-generation panel methodological approach specifically for low-income countries. First, I apply Pesarans (2015, 2021) cross-sectional dependence tests to identify interdependencies across countries. Second, I employ second-generation panel unit root tests to determine the integration properties of key variables. Third, I utilize the panel cointegration tests Pedroni (1999, 2004) and Westerlund (2005) developed to establish long-run equilibrium relationships. Finally, I estimate a panel autoregressive distributed lag (ARDL) model using a unique dataset of 28 low-income countries from 1991 to 2016 to quantify both short-run and long-run effects. To enhance analytical robustness, I explicitly control for energy prices as an endogenous variable (Lee and Lee, 2010; Costantini and Martini, 2010; Belke et al., 2011)

3.1 Cross-sections dependence

The first step of my empirical analysis begins by testing for cross-sectional dependence in the panel data. Accounting for this dependence is essential when analyzing macroeconomic data across multiple countries, as the increasing economic, social, and technological interconnectionsfueled by globalizationcan generate significant cross-country linkages. Ignoring such dependence may lead to biased or inefficient estimates. To address this, I employ the Pesaran (2015) test, which effectively identifies inter-country relationships and potential spillover effects within the panel structure, thereby enhancing the robustness and credibility of the subsequent econometric results.

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$$CD = \sqrt{\frac{2T}{N(N-1)}} \left(\sum_{i=1}^{N-1} \sum_{j=i+1}^{N} \hat{\rho}_{ij} \right), \tag{1}$$

where ρ_{ij} signifies the correlation efficiency, T represents the time dimension, and N denotes countries. The null hypothesis is either strict cross-sectional independence or weak cross-sectional dependence.

3.2 Panel unit root tests

To account for potential cross-sectional dependence in the panel dataset, I employ second-generation panel unit root tests, following the guidance of Pesaran (2007). Specifically, I begin by assessing the stationarity properties of the data using the Cross-sectionally Augmented IPS (CIPS) test. This method relaxes the assumption of cross-sectional independence by incorporating cross-sectional averages into the standard ADF regressions. The CIPS test is constructed from the simple average of the individual cross-sectionally augmented Dickey-Fuller (CADF) statistics, which are asymptotically valid and robust to heterogeneous factor loadings. The CADF regression is estimated below for each cross-sectional unit to determine the order of integration for the variables under study.

$$\Delta Y_{it} = \gamma_i + \gamma_i Y_{it-1} + \gamma_i \bar{X}_{t-1} + \sum_{l=0}^{p} \gamma_{il} \Delta \bar{Y}_{t-l} + \sum_{l=1}^{p} \gamma_{il} \Delta Y_{it-l} + \mu_{it}, \tag{2}$$

where $\bar{X}_{t-1}$ and $\Delta \bar{Y}_{t-1}$ are the cross-sectional averages. Δ is the first difference operator, Y_{it} is the series of observations for country i for $t = 1, \ldots, T$ time periods. The term $\gamma_i Y_{it-1}$ captures the lagged level of the variable to detect unit roots, γ_i represents individual-specific fixed effects, and μ_{it} is the error term.

I construct the cross-sectionally augmented panel unit root (CIPS) test statistic as follows:

$$CIPS = \frac{1}{N} \sum_{i=1}^{N} CADF_i, \tag{3}$$

where $CADF_i$ indicates the ith value of the cross-sectionally augmented Dickey-Fuller test in Eq. (2). The test involves estimating the log-transformed variables at levels and first difference, with the null hypothesis being that all series are non-stationary.

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3.3 Panel cointegration

The third step of my empirical analysis involves estimating the long-run relationship between GDP, energy efficiency, and energy price. To achieve this, I apply the Pedroni (1999, 2004) panel cointegration technique, which accounts for heterogeneity across panel members and mitigates cross-sectional dependence by demeaning each series to remove common shocks (Levin et al., 2002). This method represents an improvement over traditional panel cointegration approaches such as Kao (1999), as it allows for heterogeneous cointegrating vectors and individual-specific deterministic trends across panel members. Following Pedroni's framework, I specify the estimated cointegrating relationship as follows:

$$LGDP_{it} = \alpha_{it} + \delta_{it} + \beta_1 LETY_{it} + \beta_2 LECPI_{it} + \varepsilon_{it}, \tag{4}$$

where $LGDP$, $LETY$, and $LECPI$ represent the observable variables of gross domestic product, energy intensity, and energy price (all in log form). Here, $t = 1, \ldots, T$ represents time periods, and $i = 1, \ldots, N$ are the panel members. The term α_i denotes country-specific effects, δ_t is an individual-specific deterministic time trend, and ε_{it} is the residual, capturing deviations from the long-run equilibrium. Under Pedronis residual-based panel cointegration approach, I assess the presence of cointegration by testing for unit roots in the estimated residuals. I begin by estimating the residuals from the hypothesized long-run relationship and then test whether they exhibit stationarity. The null hypothesis posits no cointegration, while the alternative indicates cointegration across all panels, either with a common or country-specific autoregressive parameter. To conduct this test, I fit the following regression model to the estimated residuals:

$$\hat{\varepsilon}_{it} = \gamma_i \hat{\varepsilon}_{it-1} + \mu_{it}, \tag{5}$$

where $\hat{\varepsilon}_{it}$ represents the estimated residuals after fitting the model in Equation (4). Rejecting the null hypothesis confirms that the system exhibits a stable long-run equilibrium between energy efficiency, GDP, and energy prices. Given the presence of cross-sectional dependence, the Pedroni (2004) test is particularly suitable for our study as it accounts for common global shocks, ensuring robust inference in a panel setting.

Similarly, I apply the Westerlund (2005) cointegration test by examining the stationarity of the error correction term. The test evaluates whether the residuals of the long-run relationship are stationary (I(0)), which would indicate the presence of cointegration. The null hypothesis states that no cointegration exists among the variables in the panel, while the alternative suggests that cointegration is present in at least some panel members. I rely on the Variance Ratio (VR) statistic, which accounts for cross-sectional dependence and heterogeneity, and tests the null hypothesis of no cointegration against the alternative that cointegration exists for at least some panel members.

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3.4. Panel cross-sectional autoregressive distributed lag model (CS-ARDL)

To examine the dynamic relationships among energy efficiency, economic growth, and energy prices, I employ the panel Autoregressive Distributed Lag (ARDL) framework developed by Pesaran and Shin (1997) and Pesaran et al. (2001). This methodological approach is particularly appropriate for panels with variables integrated of different orders, as established by Pesaran et al. (1999). For robustness, I estimate three complementary panel ARDL specifications: Mean Group (MG), Pooled Mean Group (PMG), and Dynamic Fixed Effects (DFE). Subsequent Hausman test results support the selection of the PMG estimator, which imposes long-run homogeneity while accommodating heterogeneous short-run dynamics across countries. This specification efficiently addresses serial correlation and endogeneity concerns while enhancing estimation efficiency in panel settings. To account for cross-sectional dependence, a critical consideration in multi-country panels, I further implement the Cross-Sectionally Augmented ARDL (CS-ARDL) model following recent methodological applications by Xu et al. (2024) and Deka et al. (2023).[7]

Long run interpretation. The long run relationship appears inside the error correction term, $[LGDP_{i,t-1} - \lambda_i' X_{it}]$ and it defines the equilibrium link between gross domestic product and the explanatory variables.The vector λ_i gives the long run elasticities, and when the term equals zero, output aligns with its equilibrium value. Otherwise (any nonzero value) signals a deviation from the long run path.

Short run interpretation. The short run dynamics enter through the differenced variables. The coefficients on $\Delta LGDP_{i,t-j}$ and $\Delta X_{i,t-j}$ show how output responds to short run movements in its own past changes and in the regressors.These parameters capture immediate adjustments that occur before the system returns to its long run equilibrium. The cross sectional averages $\bar{Z}_{t-j}$ account for common shocks that affect all units in the sample. The model is specified below:

$$\Delta LGDP_{it} = \theta_i \left[LGDP_{i,t-1} - \lambda_i' X_{it}\right] + \sum_{j=1}^{\rho-1} \xi_{ij} \Delta LGDP_{i,t-j} + \sum_{j=0}^{q-1} \beta_{ij}' \Delta X_{i,t-j} + \sum_{j=0}^{r} \tau_{ij}' \bar{Z}_{t-j} + \phi_i + \varepsilon_{it} \tag{6}$$

[7]The use of the CS-ARDL estimator in this study has a number of attractive advantages. First, it addresses the endogeneity problem by estimating both short-run and long-run coefficients simultaneously, incorporating lagged dependent and explanatory variables, as well as cross-sectional averages. As highlighted by Pesaran and Shin (1999), selecting appropriate lag orders enables the ARDL model to correct for both serial correlation and endogeneity. In addition, Jalil et al. (2013) further noted that if the estimated ARDL model is free of serial correlation, endogeneity becomes less problematic. In this approach, all variables are treated as endogenous, and both short-run and long-run parameters are estimated jointly (Khan et al., 2021). Second, the ARDL approach is robust in small samples and produces coefficient estimates that are super-consistent. Finally, it is applicable irrespective of whether the underlying regressors are purely or mutually cointegrated, thereby avoiding the potential bias associated with unit root and cointegration tests.

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where:

$$\begin{aligned}
\theta_i &= -(1-\delta_i), \quad \text{group-specific speed of adjustment coefficient,} \\
\lambda_i' &= \text{vector of long-run relationships,} \\
\text{ECT} &= \left[LGDP_{i,t-1} - \lambda_i' X_{i,t}\right], \quad \text{the error correction term,} \\
\xi_{ij}, \beta_{ij}' &= \text{short-run dynamic coefficients,} \\
\overline{Z}_{t-j} &= \text{cross-sectional averages of the dependent and independent variables.}
\end{aligned}$$

4. Data and Country Clustering

I focus exclusively on low-income countries, whereas previous studies often aggregate different income groups without isolating low-income dynamics. By classifying countries based on wealth, energy efficiency, resource endowment, and economic structure, I capture intra-group heterogeneitya dimension frequently overlooked in existing literature. While methodologically aligned with other second-generation panel studies, this analysis reveals how structural constraints and institutional weaknesses distinctively shape the energy efficiencyeconomic growth nexus in low-income economies.

I utilize pooled annual time series data on energy efficiency (ETY), gross domestic product (GDP), and energy prices (ECPI) for 28 low-income countries over the period 1991 to 2016. I extract data on total final energy consumption and GDP from the World Bank's Development Indicators (WDI) database, while energy price data come from Ha, Kose, and Ohnsorge (2023), who developed a comprehensive dataset titled *"One-Stop Source: A Global Database of Inflation"* (Policy Research Working Paper No. 9737, World Bank, Washington, DC).[8]The primary explanatory variable is energy efficiency,[9]which serves as an indicator of the amount of energy required to produce a single unit of economic output. While this measure captures the productivity dimension of energy use, other measures exist in the literature. Common alternatives include energy intensity, energy productivity, sector-specific efficiency indicators (for example, industrial energy use per unit of output), and physical efficiency metrics such as conversion efficiency or technical efficiency scores derived from Data Envelopment Analysis. These alternatives highlight different aspects of how economies use energy but often reflect similar underlying relationships when applied at the macroeconomic level. To further understand the dynamics of EE on growth amidst the heterogenities present in LICs, I categorize the 28 countries into four clusters based on selected economic and energy-related indicators such as wealth, energy efficiency, resource endowments, and economic structural uniqueness(drawn from the IMF's classification of LICs).This IMF framework provides a coherent and policy-relevant basis for grouping countries by structural characteristics, ensuring

[8]The dataset on energy inflation is accessible at https://www.worldbank.org/en/research/brief/inflation-database.
[9]I measure energy efficiency as GDP per unit of total final energy consumption—a standard proxy that reflects how much economic output is generated per unit of energy used

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comparability while capturing heterogeneity in economic structure, fiscal capacity, governance, institutional strength, and resilience. Given that energy efficiency outcomes are influenced by these same structural factors, the IMF framework provides the most appropriate baseline for empirical analysis in this study. Finally, the scope of this analysis is determined by the availability of consistent time series data across all selected countries. Below is the complete list of countries included in this study.

- Benin
- Burkina Faso
- Burundi
- Cabo Verde
- Cameroon
- Chad
- Congo
- Côte d'Ivoire
- Ethiopia
- Ghana
- Grenada
- Guinea
- Honduras
- Kenya
- Kyrgyzstan
- Lesotho
- Maldives
- Mali
- Mozambique
- Myanmar
- Nepal
- Senegal
- Saint Lucia
- St. Vincent and the Grenadines
- Tajikistan
- Togo
- Tonga
- Zambia

5 Empirical Results

In this section, I present the estimation results of the relationship between energy efficiency and economic growth across 28 low-income countries using a four-stage methodological approach. The cross-sectional dependence tests reveal significant interconnections among sample economies, justifying the use of second-generation panel techniques. Panel unit root tests show mixed integration orders, while cointegration analysis confirms long-run equilibrium relationships between the variables in most country clusters. The Cross-Sectional Pooled Mean Group ARDL estimation reveals that while energy efficiency generally yields positive long-run economic benefits in the full sample, substantial heterogeneity exists across clusterswith high-efficiency economies experiencing diminishing returns and less developed economies struggling to translate efficiency gains into growth due to structural constraints and adjustment costs.

5.1 Cross-Sectional Dependence

The Pesaran CD test results decisively reject the null hypothesis of weak cross-sectional dependence at the 1% significance level across all variables. This finding reveals substantial cross-sectional dependence in the residuals throughout the panel countries, indicating that economic shocks originating in one country transcend domestic boundaries and propagate across borders. This evidence calls

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Table 1: Cross-Sectional Dependence Test Results

Variable	CD-test	p-value	Corr	abs(Corr)
Energy Efficiency	13.89	0.000	0.140	0.570
Gross Domestic Product	90.67	0.000	0.915	0.915
Energy Price	8.46	0.000	0.085	0.209

for coordinated regional energy and economic policies, as isolated interventions prove ineffective in interconnected environments. Instability in one economystemming from energy disruptions, price fluctuations, or macroeconomic volatilitycan undermine efficiency initiatives and growth in neighboring states, particularly in low-income countries with fragile institutions.[10]Consequently, while cross-sectional dependence appears across diverse income classifications, its underlying mechanisms and policy implications reveal distinctively within low-income economic contexts.

5.2. Panel unit root results.

Table 2: Panel Unit Root Test Results (CIPS)

	Full Sample		Wealthiest		Resource-rich		Low Efficiency		Unique Economies	
	Level	Diff.	Level	Diff.	Level	Diff.	Level	Diff.	Level	Diff.
Energy Efficiency	-1.4283	-5.1634***	-1.4559	-5.5711***	-1.5114	-5.4583***	-1.2074	-4.6477***	-1.1761	-5.4040***
Gross Domestic Product	-1.4191	-4.0297***	-2.4248**	-4.0773***	-3.0468***	-3.3186***	-1.2425	-4.7875***	-0.5764	-3.5468***
Energy Price	-3.9862***	-7.5145***	-2.4970***	-5.5661***	-3.5574***	-6.7894***	-4.9324***	-8.8105***	-4.2981***	-7.6844***

Based on Equations (2) and (3), Table 2 presents Cross-sectionally Augmented Im-Pesaran-Shin (CIPS) panel unit root test results for energy efficiency (LETY), GDP (LGDP), and energy price (LECPI). At the levels, energy efficiency exhibits non-stationarity across all clusters, while GDP shows stationarity only in the wealthiest and resource-rich countries. Energy prices are stationary at the 1% significance level across all samples, indicating an $I(0)$; process. After first differencing, all variables reject the unit root hypothesis at 1% significance, confirming mixed integration orders: energy efficiency is $I(1)$; energy prices are $I(0)$; and GDP predominantly displays $I(1)$; characteristics. This aligns with existing literature, where GDP and energy indicators typically exhibit persistent trends due to structural growth factors (Rajbhandari and Zhang, 2018). Unlike most previous studies, this research distinctively incorporates energy prices as an explicit variable in modeling the energy-growth nexus in low-income countries.

Compared to high and middle-income economies, where technological advancement has stabilized

[10]Unlike high and middle-income economies, where cross-sectional dependence emerges from advanced trade linkages and integrated markets (Xu et al., 2023; Alola et al., 2023), the interdependence among low-income countries likely results from shared structural vulnerabilities, including dependence on similar energy sources, exposure to global commodity market volatility, and constrained fiscal capacity to mitigate shocks.

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energy efficiency, (Xu et al., 2023; Ziolo et al., 2020), low-income countries show persistent non-stationarity, indicating structural weaknesses, slower clean technology adoption, and institutional barriers to energy transition. This highlights the need for long-term structural reforms in infrastructure, technology diffusion, and institution-building. While energy prices are more responsive to short-term external shocks, energy efficiency and GDP follow slow-moving dynamics requiring sustained interventions rather than relying on market adjustments alonepresenting distinctive challenges for low-income countries compared to more advanced economies.

5.3. Cross-sectional augmented Pedroni and Westerlund cointegration test results.

Table 3: Cross-Sectionally Augmented Cointegration Test Results

Sample	Statistic	Value	P-value
Full Sample	Modified Phillips-Perron	3.9406	0.0000
	Phillips-Perron	1.7989	0.0360
	Augmented Dickey-Fuller	0.8305	0.2031
	Westerlund Variance Ratio	4.3079	0.0000
Wealthiest	Modified Phillips-Perron	1.0595	0.1447
	Phillips-Perron	-0.7358	0.2309
	Augmented Dickey-Fuller	-1.6316	0.0514
	Westerlund Variance Ratio	3.6952	0.0001
Resource rich	Modified Phillips-Perron	1.0481	0.1473
	Phillips-Perron	0.2684	0.3942
	Augmented Dickey-Fuller	-1.0948	0.1368
	Westerlund Variance Ratio	0.8918	0.1863
Low efficiency	Modified Phillips-Perron	2.7012	0.0035
	Phillips-Perron	2.1736	0.0149
	Augmented Dickey-Fuller	2.0359	0.0209
	Westerlund Variance Ratio	1.8024	0.0357
Unique economies	Modified Phillips-Perron	2.7965	0.0026
	Phillips-Perron	1.7979	0.0361
	Augmented Dickey-Fuller	1.8789	0.0301
	Westerlund Variance Ratio	3.6923	0.0001

Table 3 presents the results of the cross-sectionally augmented Pedroni and Westerlund cointegration tests, providing robust evidence regarding the long-run equilibrium relationship between energy efficiency (LETY), gross domestic product (LGDP), and energy prices (LECPI) across the full sample and within the four identified clusters. The Pedroni test employs three principal statisticsModified Phillips-Perron (Mod. PP), Phillips-Perron (PP), and Augmented Dickey-Fuller (ADF) t-testswhile the Westerlund approach utilizes the variance ratio statistic. Both methodolo-

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gies consistently reject the null hypothesis of no cointegration at conventional significance levels for the full sample, indicating that despite substantial structural heterogeneity among the sampled countries, a common long-term equilibrium path exists linking these three variables. This finding significantly extends the existing literature by demonstrating that the structural interdependence between economic activity, energy efficiency advancements, and energy market dynamics persists even in low-income economiesa critical insight largely absent from previous studies that predominantly focused on high- and middle-income countries (e.g., Deka et al., 2023) where cointegration relationships often reflected more robust institutional coordination and energy policy effectiveness.

The cluster-specific findings reveal significant heterogeneities that merit careful consideration. In both low-efficiency and unique economy clusters, all test statistics strongly confirm cointegration at the 1%5% significance level, suggesting that improvements in energy efficiency are deeply and durably intertwined with economic performance and energy price dynamics in these contexts. This implies that for these economies, enhancing energy efficiency may represent a necessary conditionrather than merely a correlatefor sustainable economic growth. Conversely, the wealthier economies cluster exhibits only weak evidence of cointegration, while the resource-rich cluster demonstrates no statistically significant cointegration relationship.

These nuanced cointegration results fundamentally challenge the 'one-size-fits-all' perspective frequently embedded in global energy-growth studies and emphasize the necessity for differentiated policy frameworks tailored to specific economic contexts. For low-efficiency economies, the strong cointegration relationship suggests that integrating energy efficiency policies directly into growth strategies may yield higher and more sustainable returns on investment. Conversely, in resource-heavy contexts, additional structural adjustments and institutional reforms may be required to forge stronger linkages between energy efficiency improvements and economic growth outcomes. Furthermore, the presence of mixed integration orders in our preliminary unit root analysis, combined with the robust evidence of cointegration across clusters, provides econometric justification for employing a panel Autoregressive Distributed Lag (ARDL) model to comprehensively analyze both the short-run dynamics and long-run equilibrium relationships among these variables in subsequent sections.

5.4. Panel ARDL regression results.

5.4.1 Full Sample

In this section, I present the main results of the baseline model after estimating Eq. (6) as described in section 3. To study the effect of energy efficiency on economic growth, the baseline model controls for energy prices and cross-sectional dependence. I present the baseline estimates in Table 4, and the standard errors for each coefficient are included in parentheses. Column (1) reports the Pooled Mean Group (PMG) results of my baseline specification for the full sample of 28 countries from

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Table 4: Pooled Mean Group Estimation Results

	Full Sample		**Wealthiest**		**Low Efficiency**		**Unique Economies**	
	LR	SR	LR	SR	LR	SR	LR	SR
ECT	0.0980**		0.00885		0.151**		0.127	
	(0.0390)		(0.00944)		(0.0757)		(0.0963)	
L.Energy Efficiency		-0.283***		-0.282***		-0.474***		-0.308***
		(0.0558)		(0.0920)		(0.124)		(0.110)
L. Energy Price		-0.00117**		-0.00154		-0.00145		0.000297
		(0.000572)		(0.00276)		(0.00173)		(0.000577)
csavg.L.Energy Efficiency		0.216**		0.269**		0.115		-0.176
		(0.0921)		(0.122)		(0.146)		(0.137)
csavg.L.Gdp		0.00989		0.0699		-0.0642		-0.0963
		(0.0528)		(0.0557)		(0.0652)		(0.110)
csavg.L.Energy Price		0.00432		0.0101		-0.000649		-0.000570
		(0.00264)		(0.00619)		(0.00455)		(0.00669)
L.Energy Efficiency	0.376***		1.111		-1.040***		0.0696	
	(0.0523)		(3.102)		(0.0782)		(0.0854)	
L. Energy Price	-0.00276		0.350		-0.00252		-0.00352	
	(0.00416)		(0.522)		(0.00737)		(0.00652)	
Constant		-0.350		1.231***		1.126		-2.566*
		(0.654)		(0.388)		(1.147)		(1.548)
Observations	700	700	150	150	200	200	200	200

Standard errors in parentheses *** p<0.01, ** p<0.05, * p<0.1. CSAVG - Cross-sectional averages. LR=Long-run, SR=Short-run.

1991 to 2016, whereas columns (2)-(4) report the results of the three chosen sub-samples, namely, the wealthiest, low-efficient, and unique economies, respectively. I highlight two main results from my estimation in Table 4. First, an increase in energy efficiency contributes positively to economic growth in the long-run but has a negative effect in the short-run. Then I find evidence of the LICs heterogeneities in promoting or hindering these positive effects. In the following paragraphs, I discuss the results in more detail[11]

In the full sample analysis, my result shows a positive and statistically significant long-run relationship between energy efficiency and economic growth at 1% significance level. These productivity gains from electricity materialize through several interrelated channels. First, it enables firms to automate production processes, optimize input combinations, and increase capacity utilization. Second, it enhances labor productivity by improving workplace conditions. Third, it supports the consistent and efficient operation of modern technologies that are essential for scaling production. Together, these mechanisms raise output per unit of input and form the foundation for the long-term growth benefits associated with energy efficiency improvements. In addition, my findings are consistent with Cantore et al. (2016), Deichmann et al. (2018), Dzwigol et al. (2023), Brlei Vali

[11] A potential limitation of CS-ARDL models is that long-run coefficients are often less precise than short-run estimates because they rely on the estimated cointegrating vector, which is sensitive to small samples, cross-country heterogeneity, and noise in slow-moving variables such as efficiency and energy prices. These factors can inflate long-run magnitudes or reduce precision. However, short-run estimates, based on differenced data, are typically more stable. Nevertheless, the robustness checks in this paper mitigate these concerns, but the long-run results should still be interpreted cautiously.

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and Markuz (2024), Xu et al. (2024), and Ahmed and Elfaki (2024), validating energy efficiencys significant role in driving long-term economic growth. However, in the short run, energy efficiency exerts a negative and statistically significant effect on economic growth, likely due to initial adjustment costs from infrastructure changes, technology investments, and industry restructuring before long-term gains materialize. In addition, Fowlie and Meeks (2021), in their structured literature review, further substantiate that in low- and middle-income countries, energy efficiency investments often underperform initially due to market failures, credit constraints, weak infrastructure, low willingness to pay, and high rebound effects. Hence, these transitional frictions precede the realization of long-term benefits.

Subsequently, the result further shows that energy price has a negative but insignificant effect on economic growth in the long run. This is unsurprisingly due to widespread government subsidies (Fowlie and Meeks (2021), as well as reliance on traditional energy sources, and external aid to cushion the impact of rising energy prices. Conversely, energy price negatively and significantly affects growth in the short run, an indication that an increase in energy price immediately hurts growth through an increase in production cost, a fall in consumer spending, and low investment. Finally, the slow adjustment to equilibrium underscores the structural rigidities, institutional weakness, and market inefficiencies present in low-income countries.

5.4.2 Clusters

Next, I investigate this further by narrowing the countries into 3 clusters or sub-samples in order to exploit the heterogeneities present in LICs, and ascertain how these economic, institutional, and structural differences contribute to or hinder the growth potentials of EE.

Column (2) of Table 4 presents the results of the group of wealthy and highly efficient LICs. The results show a positive but statistically insignificant long-run effect of energy efficiency on economic growth, but a negative and significant short-run effect. Expectedly, energy prices show no long-run effect but a negative and significant short-run effect. This result aligns with what is documented in high-income economies. For e.g., Sueyoshi and Goto (2023) and Howarth (2007) identifies strong long-run causality between energy efficiency and economic growth, although decreasing returns to scale from existing efficiency gains likely attenuate this relationships significance in more advanced economies. However, the diminished long-run effect evident in this cluster likely reflects economic maturity, market resilience, energy diversification, and government interventions.[12] During the sample period, many of these countries implemented energy-efficiency and renewable projects in Kyoto-

[12]To address the identified challenges present in the wealthy and highly efficient LICs, those countries should adopt targeted, innovation-led policies that ease transitional costs, strengthen institutional responsiveness, and prioritize advanced energy strategies such as smart grids and R&D. Enhancing market resilience and setting sector-specific energy efficiency targets can further ensure effective policy impact

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related policy frameworks, including Kenyas industrial energy-efficiency programme (2006–2012), the Maldives Renewable Energy Technology Development and Application Project (UNDP/GEF, 2007–2012), and Ghanas public-sector energy-efficiency reforms (2007–2010). Many also piloted energy-efficiency standards or engaged in carbon-creditlinked projects in industry and power. These interventions raised baseline efficiency and strengthened resilience to both energy-price and policy shocks. Additionally, the positive but insignificant error correction term indicates institutional rigidity and slow adjustment to shocks, factors typically absent in high-income studies but critical for low-income economies transitioning toward high-efficiency systems.

Furthermore, I examine countries that have low wealth and efficiency levels. From the results in column (3), it is evident that a unit change in energy efficiency significantly reduces economic growth both in the short run and long run, and the speed of adjustment is slow but positive. This is an opposite of economic theory; however, there are empirical findings that corroborate this result. In the works of Fowlie and Meeks (2021), they highlighted that low-income countries especially the poorer ones face significant barriers such as capital constraints, institutional weaknesses, significant implementation barriers, and dependence on traditional energy sources capable of undermining the anticipated productivity gains from efficiency improvements in the short and long run.[13] Empirically, many of these economies did not attract large-scale Clean Development Mechanism (CDM) or carbon-credit projects in energy efficiency. When efficiency investments occurred, such as improved cookstove programs in Mozambique and improved cookstoves and charcoal-efficiency projects in Burkina Faso, they often focused on small, donor-funded interventions in public lighting, cookstoves, or small industry, with limited spillovers to broader productivity. At the same time, the aforementioned barriers suppress output when firms and households reallocate scarce resources toward compliance or equipment upgrades. Similarly, as observed in the other sub-groups of low-income countries, a unit increase in the energy prices presents no significant impact on growth. A further confirmation to the insulation of these economies to energy price fluctuations. My results provide compelling evidence of growth-dampening effects from efficiency interventions in low-wealth, cum efficiency, and institutionally fragile countriesa critical insight for development policy prioritization.

Finally, I analyze the effect of EE on the economic growth of unique economies. The results in column (4) imply that a unit increase in EE does not significantly contribute to long-run economic growth but does affect the growth of these countries negatively in the short run. Similarly, energy price fluctuations appear to have a negligible impact on growth across both horizons. These results strongly align with the works of Rajbhandari and Zhang (2018), Go et al. (2020)[14]and Cantore,

[13]For the low wealth and efficiency levels countries, they should adopt supportive policy frameworks that prioritize capacity building, infrastructure development, and access to affordable financing. Specifically, reforms should focus on strengthening institutional quality, phasing in energy efficiency targets gradually, targeting high-inefficiency sectors such as agriculture and manufacturing, and embedding EE goals within broader national development plans.

[14]By using Panel VECM, Rajbhandari and Zhang (2018) show that in lower-middle-income economies, the aggregate energy efficiencygrowth effects often weaken once structural composition is controlled for, indicating that a diverse sectoral mix can dilute efficiency gains. They further reveal that industrial energy efficiency shows a stronger

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Cali, and te Velde (2016).[15]They highlighted that the growth impact of EE tends to be less pronounced once structural composition is controlled for. This is evident in economies characterized by structurally diverse sectors, such as tourism, remittances, and agriculture, due to the concentration of economic activities in less energy-intensive sectors. These findings suggest that governments should replace broad energy efficiency (EE) mandates with sector-specific strategies like targeted renewable-energy projects in Kyoto-era mechanisms such as CDM projects, renewable-energy investments like Nepals hydro expansion (2008), the Caribbean Hotel Energy Efficiency Action Program (2009–2013) in Saint Lucia, and, in some cases, carbon-credit sales that target key areas, such as tourism, agriculture, and services. Focused improvements in hotel energy use, cold storage for exports, and public facilities can deliver more relevant and impactful outcomes than generalized industrial EE reforms. Finally, to reduce short-run disruptions, governments should provide targeted support, such as tax incentives, soft loans, and technical assistance, to ease the transition to energy-efficient technologies in priority sectors, while strengthening institutional capacity through training, data systems, and public awareness

In conclusion, my empirical result shows that while energy efficiency fosters long-term growth at the aggregate level, it can suppress growth in the short run due to transitional frictions. Similarly, there are differences in the effects across the LICs due to the diverse and heterogeneous features inherent in those countries.

6.Robustness Analysis

6.1 Robustness to Energy Mix Composition

Differences in energy generation sources can distort standard measures of energy efficiency. Countries that rely heavily on biomass(like Benin, Burkina Faso)often appear energy-intensive because biomass has low heat output and large conversion losses, while countries with greater access to hydro or solar generation (such as Tajikistan, Zambia, Nepal etc.) tend to record lower energy intensity due to higher conversion efficiency. This implies that part of the observed cross-country variation in measured efficiency reflects differences in energy sources rather than genuine improvements in production technologies. To address this concern, I extend the analysis by incorporating indicators of electricity composition and restrict attention to 13 countries with more complete data on renewable generation and oil-, gas-, and coal-based electricity.

and more direct link to economic growth than agricultural efficiency, particularly in upper-middle-income countries. This aligns with my findings that sectoral composition plays a significant role in shaping the economic returns of energy efficiency

[15]In addition, Cantore, Calì, and te Velde (2016) highlighted the role of sectoral composition in shaping EE-economic gains with notable variation across industries such as textiles, food processing, and chemicals in the manufacturing sector. Instructively, my study contributes to the body of literature by providing compelling evidence that EE economic gain will not only be weakened by the sectoral mix of countries but can also actively impede growth in structurally constrained economies such as those found in cluster 3

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To assess whether the baseline results are sensitive to these structural differences, I augment the CS-ARDL model with the shares of renewable and oil, gas, and coal electricity production. The long-run coefficient on energy efficiency remains positive, stable, and highly significant across both models, confirming that the growth-enhancing effect of efficiency is not driven by omitted energy-mix characteristics. Introducing renewable and oil-, gas-, and coal-based electricity shares slightly increases the magnitude of the long-run efficiency effect, while the renewable share shows negative influence, suggesting that the current renewable generation mix in these LICs may be associated with unreliability or insufficient grid capacity. Oil-, gas-, and coal-based electricity shares have no statistically significant effect. Nevertheless, the short-run adjustment pattern also remains intact, with efficiency improvements negatively affecting growth in both specifications. Overall, the results demonstrate that the core findings are robust to controlling for heterogeneity in energy sources.

6.2 Reclustering by Energy Use Patterns

As an additional robustness exercise, I reclassify the 28 LICs using a k-means clustering scheme based solely on energy-system characteristics rather than economic structure. I grouped the LICs based on their average energy consumption per capita, which captures heterogeneity in technology adoption, energy infrastructure, and energy sector development that may shape the energy efficiency–growth relationship. The PMG estimates reported in Table 11 show that the main results remain stable across these alternative groupings. For low-, medium-, and high-energy users, energy efficiency continues to exhibit negative and statistically significant short-run effects, reflecting adjustment frictions, infrastructure constraints, and low absorptive capacity. In contrast, the long-run effect of energy efficiency becomes significant only in the low-energy group like Kyrgyzstan, Grenada, Honduras, Zambia, and others, where more reliable energy systems and greater integration of modern energy inputs support stronger productivity gains. These findings confirm that the core conclusions of the paper do not depend on the original classification and are robust to an alternative clustering based on energy consumption patterns.

6.3 Addressing Reverse Causality

There may be potential concerns about reverse causality, where economic growth itself influences improvements in energy efficiency. In this paper, I address the issue directly through several features of the empirical design. First, the CS-ARDL model incorporates optimal lagged values of both GDP and energy efficiency, which separates contemporaneous movements from the underlying causal mechanism. This specification forces the direction of adjustment to run from previous efficiency improvements toward current economic performance rather than the reverse. Second, the inclusion of the error-correction term captures the long-run equilibrium relationship among GDP, energy efficiency, and energy prices, ensuring that the long-run coefficients reflect the adjustment

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back to equilibrium rather than short-run joint correlation. Because the long-run parameters are estimated from the cointegrating vector, they are immune to simultaneity that typically affects short-run regressions. Third, the cross-sectional averages included in the model specification absorb common shocks, technological spillovers, and region-wide growth fluctuations that could confound the relationship between efficiency gains and GDP. These controls remove common factors that otherwise generate false causal links. Finally, by estimating heterogeneous short-run dynamics across countries, the PMG estimator avoids imposing homogeneity where causal channels differ, preventing pooled dynamics from creating artificial associations that resemble reverse causality. Together, these design choices ensure that the estimated long-run effect of energy efficiency on growth reflects genuine adjustment behavior rather than growth-induced improvements in efficiency.

Nevertheless, to further strengthen the reverse-causality argument, I estimate a dynamic System GMM model where energy efficiency is instrumented using deeper lags. This approach generates exogenous variation in efficiency and tests whether growth shocks drive measured efficiency gains. The results (Appendix Table 12) show that the efficiency coefficient becomes statistically insignificant once instrumented, indicating no evidence of forward causality from GDP to efficiency.

7 Conclusion

In this study, I investigate the dynamic relationship between energy efficiency and economic growth in 28 LICs, focusing on how heterogeneities in these countries either enhance or hinder growth. Using second-generation panel econometric methods, I present empirical evidence that while energy efficiency (EE) can support long-run economic growth, its effects are far from uniform. The overall EEgrowth nexus is positive, but heterogeneities are central to understanding how these gains materialize across the different countries. Wealthier and institutionally stronger LICs are better able to capture the long-term benefits of EE, whereas poorer or structurally unique economies may face weak or even adverse outcomes. Recognizing these differences underscores the importance of studying LICs not as a homogenous group but through the lens of their heterogeneous energy systems, economic structures, and institutional capacities. Perhaps, this perspective not only provides a more accurate empirical picture of the EEgrowth relationship but also strengthens the argument for country-specific policies towards actualizing the positive effects of EE on growth.

This study makes a significant contribution to the energy-growth literature by providing one of the first empirical analyses focused exclusively on low-income countries. More importantly, it examines their internal heterogeneities, highlighting how these differences shape the energy efficiencygrowth relationship. At the same time, my study has limitations. First, it remains constrained by data availability and focuses on aggregate country-level measures of energy intensity, which may conceal variation across regions or specific industries. Further research could integrate micro-level data, such as firm-level energy use or sectoral efficiency indicators, to unpack the mechanisms behind

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the observed outcomes. In addition, future work could extend the sample period to capture more recent transitions in renewables, incorporate environmental outcomes, and test for the moderating role of indicators like governance, institutional quality, inequality, or foreign investment.

In conclusion, my findings offer valuable policy lessons and practical implications. First, energy efficiency does not improve growth uniformly. As a result, EE reforms and policies cannot be enacted or adopted entirely in LICs without providing a supportive economic, infrastructure and institutional environment. For fragile or low-efficiency LICs, complementary reforms, such as improving grid capacity, easing capital constraints, and targeting high-use sectors such as tourism, cold storage, or irrigation, are critical. For wealthier LICs, policies should focus on sustaining efficiency without triggering diminishing returns. Second, adverse short-term effects highlight the need for policymakers to design phased reforms with transitional support (e.g., subsidies, soft loans, or training programs) to ease adjustment. Overall, given the differences in the policy prescriptions across LICs, it is pertinent to highlight that policy reforms shape countries differently. Only context-sensitive EE policies tailored to country clusters and sectoral needs can ensure that efficiency investments contribute to sustainable economic growth.

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Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

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Appendix

Table 5: Descriptive Statistics

Variable	Cluster 1	Cluster 2	Cluster 3	Cluster 4	Overall
Panel A: GDP per capita					
Mean	22.831	23.108	22.234	21.856	22.424
Std. Dev.	1.383	0.743	0.658	1.636	1.286
Between	1.445	0.756	0.573	1.693	1.261
Within	0.402	0.269	0.380	0.396	0.377
Min	19.577	21.708	20.756	19.443	19.381
Max	25.015	24.617	23.555	24.981	25.015
Panel B: Energy efficiency					
Mean	8.875	9.299	9.765	9.082	9.284
Std. Dev.	0.696	0.618	0.605	1.213	0.915
Between	0.726	0.634	0.588	1.256	0.896
Within	0.205	0.212	0.249	0.289	0.250
Min	7.534	8.016	8.627	7.170	7.107
Max	10.268	10.884	11.176	11.198	11.198
Panel C: Energy prices					
Mean	2.146	2.197	1.682	1.590	1.865
Std. Dev.	1.817	1.487	1.580	1.753	1.693
Between	1.107	0.771	0.655	0.970	0.878
Within	1.508	1.309	1.456	1.498	1.458
Min	-2.813	-2.904	-3.467	-3.679	-3.679
Max	6.293	7.601	4.809	4.809	7.601
Observations					
N	156	156	208	208	756
n (countries)	6	6	8	8	28
T (years)	26	26	26	26	27

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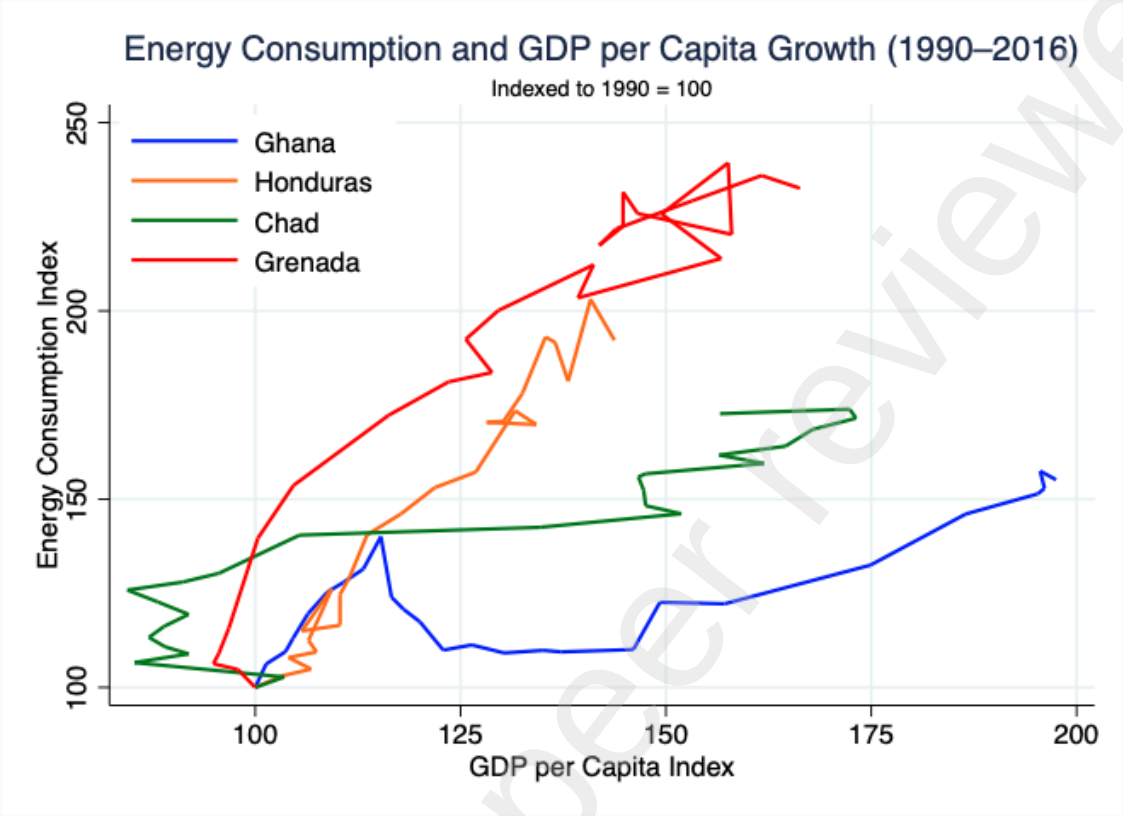


Figure 1: Trends in Energy Consumption and GDP, 1990–2016

Table 6: Comparison of PMG and DFE Estimation Results

	PMG			DFE		
	Coef.	Std. Err.	P> |z|	Coef.	Std. Err.	P> |z|
Long-run coefficients						
L.Energy efficiency	0.3764	0.0523	0.000	-1.2564	0.3090	0.000
L. Energy price	-0.0028	0.0042	0.507	-0.0669	0.0422	0.113
Error correction and short-run dynamics						
EC term	0.0980	0.0390	0.012	-0.0430	0.0128	0.001
D.L.Energy efficiency	-0.2825	0.0558	0.000	-0.1443	0.0206	0.000
D.L. Energy price	-0.0012	0.0006	0.040	0.0016	0.0012	0.175
csavg.L.Energy efficiency	0.2160	0.0921	0.019	0.3021	0.1067	0.005
csavg.L.gdp	0.0099	0.0528	0.851	0.1491	0.0383	0.000
csavg.L.Energy price	0.0043	0.0026	0.102	0.0115	0.0038	0.002
cons	-4.0609	1.6589	0.014	-4.6855	1.7964	0.009

Note: PMG = Pooled Mean Group estimator. DFE = Dynamic Fixed Effects estimator. P-values are based on robust standard errors. EC term refers to the error correction coefficient.

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Table 7: Hausman Test Comparing PMG and DFE Estimates

	PMG (b)	DFE (B)	b − B	Std. Err.
Energy Efficiency	0.3764	−1.2564	1.6328	6.9819
Energy Price	−0.0028	−0.0669	0.0641	0.5535

Note: This table reports the Hausman test comparing PMG and DFE estimators. Null hypothesis: No systematic difference between estimators. $\chi^2(2) = 0.07$, p-value = 0.9641. Fail to reject H_0, implying PMG is efficient and preferred as the baseline model.

Table 8: PMG Estimation Results by Cluster

	Cluster 1: High Efficiency & Wealth			Cluster 3: Low Efficiency & Poor			Cluster 4: Unique Economies		
	Coef.	Std.Err.	P> |z|	Coef.	Std.Err.	P> |z|	Coef.	Std.Err.	P> |z|
Long-run coefficients									
L.Energy efficiency	1.1112	3.1015	0.720	-1.0400	0.0782	0.000	0.0696	0.0854	0.415
L. Energy price	0.3497	0.5218	0.503	-0.0025	0.0074	0.733	-0.0035	0.0065	0.589
Error correction and short-run dynamics									
EC term	0.0088	0.0094	0.349	0.1515	0.0757	0.045	0.1267	0.0963	0.188
D.L.Energy efficiency	-0.2820	0.0920	0.002	-0.4738	0.1238	0.000	-0.3077	0.1101	0.005
D.L. Energy price	-0.0015	0.0028	0.577	-0.0015	0.0017	0.402	0.0003	0.0006	0.606
csavg.L.Energy efficiency	0.2688	0.1223	0.028	0.1150	0.1461	0.431	-0.1755	0.1373	0.201
csavg.L.gdp	0.0699	0.0557	0.210	-0.0642	0.0652	0.324	-0.0963	0.1098	0.381
csavg.L.Energy price	0.0101	0.0062	0.102	-0.0006	0.0046	0.887	-0.0007	0.0067	0.932
cons	-4.3398	2.3901	0.069	-1.2578	2.1772	0.563	1.0715	2.1114	0.612

Notes: PMG = Pooled Mean Group estimator. Cluster 1 includes wealthy and energy-efficient countries, Cluster 3 includes low-efficiency and poor countries, and Cluster 4 includes small island or unique economies. P-values are based on robust standard errors.

Table 9: DFE Estimation Results by Cluster

	Cluster 1: High Efficiency & Wealth			Cluster 3: Low Efficiency & Poor			Cluster 4: Unique Economies		
	Coef.	Std.Err.	P> |z|	Coef.	Std.Err.	P> |z|	Coef.	Std.Err.	P> |z|
Long-run coefficients									
L.Energy efficiency	-0.9853	0.5849	0.092	-0.7852	0.2392	0.001	-1.1734	0.1356	0.000
L. Energy price	-0.0476	0.0575	0.407	0.0053	0.0416	0.900	0.0246	0.0241	0.307
Error correction and short-run dynamics									
EC term	-0.0491	0.0276	0.075	0.0654	0.0259	0.011	-0.1357	0.0293	0.000
D.L.Energy efficiency	-0.1355	0.0414	0.001	-0.2856	0.0346	0.000	-0.1318	0.0474	0.005
D.L. Energy price	-0.0013	0.0024	0.603	-0.0008	0.0014	0.673	-0.0008	0.0024	0.736
csavg.L.Energy efficiency	0.3720	0.1965	0.058	0.2029	0.1808	0.262	0.1833	0.2195	0.404
csavg.L.gdp	0.1752	0.0746	0.019	0.0356	0.0676	0.599	0.1459	0.0778	0.061
csavg.L.Energy price	-0.0139	0.0067	0.045	0.0095	0.0062	0.125	0.0106	0.0049	0.211
cons	-5.8049	3.3136	0.080	-4.6172	2.9940	0.123	-0.5525	3.6962	0.881

Notes: DFE = Dynamic Fixed Effects estimator. Cluster 1 = wealthy and energy-efficient countries. Cluster 3 = low-efficiency and poor countries. Cluster 4 = small island or unique economies. P-values are based on robust standard errors.

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Table 10: PMG Estimates for Selected Countries: Baseline vs Expanded Model

	Baseline Model	**With Different Energy Sources**
Long-Run Coefficients		
L.Energy efficiency	0.392 (0.057)***	0.436 (0.064)***
L.Energy price	−0.004 (0.004)	−0.003 (0.004)
L.Renewable Sources	—	−0.080 (0.023)***
L.Oil,Gas, and Coal Sources	—	−0.008 (0.009)
Short-Run Coefficients		
EC term	0.085 (0.087)	0.176 (0.128)
D.L.Energy efficiency	−0.315 (0.073)***	−0.188 (0.086)**
D.L.Energy price	−0.0015 (0.0009)	−0.0029 (0.0022)
D.L.Renewable Sources	—	−0.076 (0.105)
D.L.Oil,Gas, and Coal Sources	—	−0.023 (0.030)
csavg.L.Energy efficiency	0.419 (0.103)***	0.533 (0.244)**
csavg.L.gdp	0.124 (0.084)	0.080 (0.100)
csavgL.Energy price	0.0058 (0.004)	0.0056 (0.003)**
csavgL.Renewable Sources	—	0.028 (0.035)
csavgL.Oil,Gas, and Coal Sources	—	0.019 (0.018)
Constant	−8.32 (1.80)***	−9.88 (4.72)**
Observations	312	312
Countries	13	13

Table 11: Robustness Check: PMG Estimates by Energy Consumption Per Capita

	Low Energy Users	**Medium Energy Users**	**High Energy Users**
Long-run coefficients			
L.Energy efficiency	0.3999***	-3.7586	-0.0518***
L.Energy price	-0.00539	-0.8338	-0.00621
Short-run dynamics			
ECT term	0.0863	-0.00398	0.2016*
D.L.Energy efficiency	-0.2772***	-0.2571**	-0.3567**
D.L.Energy price	-0.00107*	0.00126	-0.00155
Cross-sectional averages			
csavg.L.Energy efficiency	0.2582*	-0.00317	0.1983*
csavg.L.gdp	0.0394	-0.0229	-0.1593
csavg.L.Energy price	0.00503	-0.00651	0.00378
Constant	-4.9126*	0.8809	-2.9246
Observations	350	200	150
Countries	14	8	6

Notes: PMG estimator with CS-ARDL specification. Standard errors in parentheses. ECT = error-correction term. *** p<0.01, ** p<0.05, * p<0.10.

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Table 12: Dynamic Panel System GMM Estimates

	Coefficient	Std. Error	p-value
Dependent variable: $\log(gdp)$			
L1.L.gdp	1.0294	0.0088	0.000
L.Energy efficiency	−0.0258	0.0218	0.246
L.Energy price	0.0114	0.0048	0.023
Constant	−0.4084	0.2419	0.103
Model Diagnostics			
Observations	492		
Countries	28		
Number of instruments	28		
AR(1) test p-value	0.002		
AR(2) test p-value	0.119		
Hansen test (p-value)	0.503		
Sargan test (p-value)	0.000		
Diff-in-Hansen: GMM levels (p)	0.135		
Diff-in-Hansen: GMM diffs (p)	0.210		
Diff-in-Hansen: IV years (p)	0.444		

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